Examining the bidirectional relationship between sleep and physical activity using pooled individual data

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Summary

Objectives: To examine the association between device-measured physical activity and sleep across the lifespan using individual participant data (IPD) meta-analysis. We furthed aimed to examine how the association differed by age.

Design: Pooled IPD analysis of accelerometer-based physical activity and sleep data from multiple observational studies.

Setting: We sourced data from 20 observational studies conducted multiple geographical regions and across the lifespan.

Participants: The dataset included 220,988 observations from 28,824 participants aged 2 to 94 years. Participants were selected from studies that used research-grade accelerometers with at least four valid days of wear time. Exclusion criteria included insufficient wear time and missing key covariate data.

Main Outcome Measures: Sleep outcomes included total sleep time, sleep efficiency, sleep onset timing, and sleep regularity. Physical activity was assessed in terms of volume (average acceleration) and intensity (intensity gradient). Mixed-effects models were used to examine associations, adjusting for age, sex, body mass index, and socioeconomic status.

Results: Higher physical activity volume was associated with longer sleep duration, greater sleep efficiency, earlier sleep onset, and more regular sleep. The association between physical activity volume and sleep regularity was the strongest and declined only slightly with age (standardised β = 0.30 [0.29, 0.31] at age 11 and 0.27 [0.25, 0.29] at age 65), whereas associations with physical activity intensity weakened with age. Conversely, more regular sleep was associated with more physical activity the following day, although this association was modest and weakened with age, and associations of sleep onset, duration, and efficiency with next-day physical activity were small.

Conclusions: Physical activity is positively associated with sleep regularity across the lifespan, suggesting it may be a useful target for interventions that aim to support consistent sleep patterns. Public health recommendations should emphasize the importance of regular physical activity for sleep health. Future research should investigate mechanisms underlying these associations and assess targeted intervention strategies.

Trial Registration: The study protocol was pre-registered on the Open Science Framework (https://osf.io/ahd96).

Introduction

Sleep is an important determinant of population health across the lifespan:[1,2] poor sleep is associated with all-cause mortality,[3] cardiovascular disease,[4] cancers,[5] and poor mental health.[6] Yet 14-68% of children and adolescents[7] and 31% of adults[8] sleep less than recommended,[9,10] and safe, accessible strategies to promote sleep are needed.[11,12] In adults, physical activity appears to improve both the quantity and quality of sleep,[13–15] and WHO recommends it for this purpose.[16] In children and adolescents, the evidence is less clear.[12,17,18] Trials suggest exercise improves sleep quality and efficiency but not total sleep time,[19] and guidelines do not support a role for physical activity in improving sleep in this age group.

The existing evidence has three main limitations. First, most studies relied on self-reported sleep and physical activity,[12,17,18] which could bias associations and limit the precision of dose-response analyses. Second, existing reviews and meta-analyses[12] pool study-level effects, which limits how well they can examine the influence of age.[20] Third, no study has examined day-to-day data in a large, geographically diverse sample spanning the lifespan. Daily associations have been synthesised separately in adults[21] and in children and youth,[22] and consumer-wearable studies with millions of person-nights included only adults and did not examine sleep regularity.[23,24] Day-level data matter because the relationship may run in both directions. Physical activity may promote sleep through circadian and stress-related pathways,[25] while a late or poor night's sleep may reduce activity the next day.[21,26] Weekly averages may mask these patterns, and pooling device-measured individual participant data can overcome these limitations.[27]

We harmonised individual participant data from accelerometer studies around the world to investigate the relationship between device-measured physical activity and sleep across the lifespan. We addressed three research questions (RQs): (RQ1) do individuals who are more physically active have better sleep duration and quality; (RQ2) does age moderate the association between physical activity and sleep; and (RQ3) is the association bidirectional?

Methods

We pre-registered all methods and analyses on the Open Science Framework (OSF) before data acquisition began (https://osf.io/ahd96).

Data sources and data acquisition

We included studies of any design that used research-grade accelerometers of any brand, worn at any location (hip, wrist, or thigh), from which raw accelerations (m/s^2 or mg) could be obtained, with a 24-hour wear protocol over at least four days, including one weekend day.

We identified potentially relevant studies by examining the reference lists of recent systematic reviews that explored the relationship between sleep and physical activity in children, adolescents, and adults.[13,17,18] We then used forward searching on potentially eligible studies.[28] We additionally performed a literature search to identify accelerometry consortiums and individual studies, searched trial registries (the Australian New Zealand Clinical Trials Registry, the World Health Organization International Clinical Trials Registry Platform, and the ISRCTN registry) and the Open Science Framework for unpublished results, and asked collaborators to help identify any additional studies.

To keep data acquisition manageable, we invited only studies with more than 400 participants, emailing chief investigators or consortium leads to request deidentified individual data. All

studies included in the pooled data meta-analysis had to have been approved by their local or institutional ethical committees, and all participants (or, for children, their parents or guardians) had to have provided written informed consent, with children providing assent where required. The lead university's Research Ethics Committee granted permission for each study's data to be obtained, and investigators transferred their deidentified data via a secure file-sharing system.

Data harmonisation and outcomes

We harmonised all accelerometer data by reprocessing the raw files with the open-source *GGIR* package (version 2.5-1) in R,[29,30] producing outcomes that could be compared across studies, device wear locations, and populations.[31] Processing used our *sleepIPD* R package, which fixed all settings across studies and documents the decisions made for specific devices.[32] The core research team or the data contributors ran *sleepIPD* on each study's raw files. *GGIR*'s sleep detection has been evaluated against polysomnography in both adults[30] and children.[33,34] Non-wear time and abnormally high accelerations were detected with the default settings,[35] classified as missing, and imputed from the same time interval on the remaining days of the week.[29]

Each day ran from waking to waking the following day, so each day's physical activity, calculated over its waking hours, was paired with the sleep period that followed it. For wrist-worn devices, the sleep period was detected with the HDCZA algorithm[30] and for hip-worn devices *GGIR* instead detected time in bed from the device's angle. For sleep, *GGIR* produced total sleep time (minutes), sleep efficiency (percentage of the sleep period, from sleep onset to final waking, spent asleep), sleep onset (clock time), and sleep regularity (the sleep regularity index: the likelihood of being in the same sleep or wake state at time points 24 hours apart, calculated for each pair of consecutive days, with higher values indicating more regular sleep). For physical activity, we used two metrics proposed by Rowlands and colleagues:[36] average acceleration (mg) —a measure of volume—and the intensity gradient (unitless), the slope of log time against log intensity, which describes how time is distributed across intensities, with less negative values indicating relatively more time at higher intensities. These metrics capture the whole intensity continuum[37] without the population-specific cut-points that collapse it into categories.[38]

Covariates and moderators

The following individual participant data were obtained when available: age, sex, height, weight, waist circumference, maturational status, socioeconomic status (SES), ethnicity, screen time, sleep medications, and medically diagnosed conditions known to affect sleep. The country, city, time zone, daylight length, and season in which data were collected were also obtained. Because studies measured socioeconomic status differently (e.g., education or income), we harmonised each study's measure into low, medium, and high categories (appendix).

Body mass index (BMI) was converted to an age- and sex-standardised z-score. For participants under 18 years we used the CDC 2000 BMI-for-age growth reference, applied with the *childsds* package. For adults we standardised BMI against the adults in this sample.

Statistical analysis

To reduce the risk of bias, we used a blind analysis approach:[39,40] we developed and tested models on a 10% holdout subsample of the dataset and conducted the final analysis once we believed the code was ready. Data screening had four stages: values outside fixed physiological limits (for average acceleration, above 200 mg); all physical activity variables from files whose mean average acceleration exceeded that limit, indicating a likely device failure (76 participants);

values more than 4 SD from the mean of their own study, so that between-study differences in device, protocol, and population were not mistaken for implausible measurement; and, for participants under 18 years, heights, weights, and BMI values that the CDC 2000 growth reference classifies as biologically implausible.

A day was valid if it recorded more than 10 hours of wear time and, where sleep was detected, a sleep period longer than 200 minutes. Participants required at least 4 valid days. A sensitivity analysis restricted the sample to the 14,981 participants with at least seven valid days.

We assumed missing covariate and moderator data to be missing at random or completely at random and imputed them using multilevel predictive mean matching, clustered by participant, with multiple imputation by chained equations in the *mice* package,[41] creating 50 imputed datasets. The imputation model also included the sleep and physical activity variables, but each analysis used only the person-days on which its outcome, its sleep or physical activity predictor, and its moderator were observed, so imputed values are only covariates. We pooled results across imputed datasets with Rubin's rules, computing 95% CIs and p values on the Barnard-Rubin degrees of freedom to account for the additional uncertainty introduced by imputation, and used $p<0.05$ to assess statistical significance. We fitted 16 models for each moderator and report all of them without adjustment for multiple comparisons, so p values close to 0.05 should be interpreted with caution. We used R (version 4.6.1) for all data cleaning and analysis.

To examine the relationship between sleep and physical activity we ran mixed-effects models with random intercepts for study and participant, accounting for the nesting of days within participants and of participants within studies.[42] We predicted each sleep indicator from each physical activity indicator, adjusting for SES, sex, and BMI z-score, with quadratic terms to allow curvilinear associations. We interacted each predictor with age to examine how associations change over the lifespan (RQ2). Age was centred at its mean across person-days (35.6 years) and scaled per 10 years. For interpretability, and we report associations at ages 11, 18, 35, and 65 years. Models predicting physical activity from the preceding night's sleep (RQ3) used each sleep indicator lagged by one day, so person-days whose preceding day was absent or ineligible did not contribute, leaving 136,319 of 168,613 eligible person-days. Sleep regularity was missing most often, so its models with physical activity volume used 125,155 person-days (RQ1) and 109,033 (RQ3).

Protocol deviations

This paper addresses the first two hypotheses in our registered protocol: that physical activity volume and intensity would be positively associated with optimum sleep duration and quality, with an inverted U-shaped dose-response, and that associations would be stronger in adolescents and adults than in children and older adults. Moderation by the other pre-registered moderators is reported in the appendix, and the remaining hypotheses will be addressed in future research. While working with the holdout data we made three changes to the protocol. We modelled study as a random rather than a fixed effect, so that estimates would generalise beyond the included studies. Sensitivity analyses of fixed-effect models gave qualitatively similar results. We analysed each sleep metric separately, because no validated composite sleep indicator could be estimated from our data. We did not log-transform skewed variables, because back-transformed predictions followed extreme exponential patterns near the edges of the data range. Log-scale models are reported in the appendix. Additionally, after running the full analysis we

identified a mistake in the BMI variable and in the imputation model, which we corrected and reran the analysis.

Results

The pooled dataset contained 220,988 days of physical activity and sleep data from 28,824 participants in 20 studies. We excluded 52,375 days with insufficient wear time, which removed 1,175 participants entirely, and a further 413 participants (619 days) with fewer than 4 valid days, leaving 27,236 participants and 167,994 person-days (Table 1; study characteristics are in the appendix). Participants were aged 2 to 94 years and 50.3% were female. Most came from Europe and South America, followed by Oceania, and the sample skewed towards younger children (2-11 years, 50.9%) and older adults (66+ years, 22.3%).

Table 1: Participant characteristics

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Numeric variables					
N	13,820	1,923	310	5,044	6,067
Age	8.5 (2.8)	15.7 (1.7)	29.2 (5.1)	57.7 (7.4)	72.7 (4.5)
BMI	17.7 (3.0)	21.6 (3.8)	26.5 (6.5)	26.8 (4.8)	27.1 (4.3)
BMI z-score	0.31 (1.10)	0.09 (1.02)	−0.07 (1.40)	−0.01 (1.03)	0.05 (0.93)
Valid Weartime Hours	23.4 (1.3)	23.4 (1.5)	23.3 (1.7)	23.8 (0.7)	23.9 (0.4)
Valid Weartime Days	5.7 (1.8)	5.0 (1.3)	5.0 (1.4)	7.5 (2.3)	6.6 (2.1)
PA Intensity Gradient (unitless)	−2.10 (0.18)	−2.23 (0.19)	−2.43 (0.19)	−2.50 (0.19)	−2.68 (0.22)
PA Volume (average acceleration in mg)	49.0 (26.4)	46.7 (14.3)	39.3 (10.6)	38.8 (11.6)	30.9 (9.7)
Sleep Duration (min)	471.7 (61.5)	416.4 (55.0)	421.0 (64.0)	409.1 (55.8)	404.3 (63.9)
Sleep Efficiency (%)	80.9 (7.0)	87.1 (6.3)	87.4 (6.2)	87.8 (5.5)	86.4 (6.2)
Sleep Onset (clock time; SD in h:mm)	21:32 (01:42)	23:29 (01:27)	23:33 (01:49)	23:25 (01:17)	23:57 (01:16)
Sleep Regularity Index	53.4 (12.2)	57.0 (12.1)	55.8 (12.8)	59.4 (11.4)	54.5 (12.5)
Accelerometer Wear Location					
Wrist	5,509 (39.9%)	1,885 (98.0%)	310 (100.0%)	5,044 (100.0%)	6,067 (100.0%)

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Hip	8,311 (60.1%)	38 (2.0%)	-	-	-
Country					
Australia	1,510 (10.9%)	305 (15.9%)	110 (35.5%)	188 (3.7%)	-
Finland	508 (3.7%)	-	-	801 (15.9%)	-
New Zealand	455 (3.3%)	-	37 (11.9%)	152 (3.0%)	-
Spain	809 (5.9%)	-	-	37 (0.7%)	2,386 (39.3%)
Switzerland	-	-	-	1,905 (37.8%)	1,023 (16.9%)
United Kingdom	447 (3.2%)	-	144 (46.5%)	1,961 (38.9%)	2,658 (43.8%)
Brazil	3,425 (24.8%)	727 (37.8%)	10 (3.2%)	-	-
Czech Republic	281 (2.0%)	848 (44.1%)	9 (2.9%)	-	-
Canada	522 (3.8%)	-	-	-	-
Chile	138 (1.0%)	5 (0.3%)	-	-	-
China	125 (0.9%)	-	-	-	-
Colombia	889 (6.4%)	-	-	-	-
Denmark	689 (5.0%)	-	-	-	-
Ecuador	981 (7.1%)	38 (2.0%)	-	-	-
India	575 (4.2%)	-	-	-	-
Kenya	528 (3.8%)	-	-	-	-
Portugal	702 (5.1%)	-	-	-	-

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
South Africa	507 (3.7%)	-	-	-	-
United States	729 (5.3%)	-	-	-	-
Ethnicity					
Non-White	2,672 (19.3%)	169 (8.8%)	14 (4.5%)	94 (1.9%)	240 (4.0%)
Other or unknown	4,780 (34.6%)	1,194 (62.1%)	154 (49.7%)	1,340 (26.6%)	2,424 (40.0%)
White	6,368 (46.1%)	560 (29.1%)	142 (45.8%)	3,610 (71.6%)	3,403 (56.1%)
Region					
Europe	3,436 (24.9%)	848 (44.1%)	153 (49.4%)	4,704 (93.3%)	6,067 (100.0%)
Oceania	1,965 (14.2%)	305 (15.9%)	147 (47.4%)	340 (6.7%)	-
South America	5,433 (39.3%)	770 (40.0%)	10 (3.2%)	-	-
Africa	1,035 (7.5%)	-	-	-	-
Asia	700 (5.1%)	-	-	-	-
North America	1,251 (9.1%)	-	-	-	-
Season					
Autumn	4,672 (33.8%)	445 (23.1%)	39 (12.6%)	1,241 (24.6%)	1,268 (20.9%)
Spring	2,859 (20.7%)	1,230 (64.0%)	125 (40.3%)	1,226 (24.3%)	1,988 (32.8%)
Summer	2,113 (15.3%)	82 (4.3%)	85 (27.4%)	1,271 (25.2%)	1,252 (20.6%)
Winter	4,176 (30.2%)	166 (8.6%)	61 (19.7%)	1,306 (25.9%)	1,559 (25.7%)
Sex					

	Age group				
Characteristic	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Female	7,154 (51.8%)	1,002 (52.1%)	222 (71.6%)	2,717 (53.9%)	2,551 (42.0%)
Male	6,664 (48.2%)	921 (47.9%)	88 (28.4%)	2,327 (46.1%)	3,516 (58.0%)
Sleep Conditions Reported					
No	625 (4.5%)	-	-	3,512 (69.6%)	5,141 (84.7%)
Yes	50 (0.4%)	-	-	310 (6.1%)	626 (10.3%)
Not Reported	13,145 (95.1%)	1,923 (100.0%)	310 (100.0%)	1,222 (24.2%)	300 (4.9%)
Socioeconomic Status					
Low	2,892 (24.1%)	208 (11.5%)	2 (0.7%)	1,074 (22.0%)	2,388 (39.4%)
Medium	4,638 (38.7%)	885 (49.1%)	191 (70.2%)	2,409 (49.3%)	2,325 (38.3%)
High	4,456 (37.2%)	708 (39.3%)	79 (29.0%)	1,406 (28.8%)	1,352 (22.3%)

For categorical variables the value is the count, and percentage. For numeric variables the value is the Mean and SD. Sleep onset is a clock time; its SD is a duration in hours:minutes. Total N = 27,236. 72 participants had missing age data and are counted in the total but not in any age band.

Physical activity volume and sleep

Higher physical activity volume was associated with longer sleep duration, higher sleep efficiency, earlier sleep onset, and more regular sleep (Table 2; Figure 1). Curvilinear terms were statistically significant but did not meaningfully change these associations. The association with sleep regularity was the largest and declined only slightly with age (β = 0.30 [0.29, 0.31] at age 11 and 0.27 [0.25, 0.29] at age 65). The associations with sleep duration and efficiency weakened with age, and by age 65 higher volume was associated with slightly lower sleep efficiency (β = −0.03 [−0.04, −0.02]), whereas the association with earlier sleep onset strengthened with age (β = −0.09 [−0.10, −0.08] at age 11 and −0.15 [−0.16, −0.14] at age 65).

Table 2: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [−0.16, 0.16]	.996	0.00 [−0.15, 0.14]	.995
Physical activity	0.10 [0.09, 0.11]	< .001	0.05 [0.05, 0.06]	< .001
Age	−0.02 [−0.03, 0.00]	.007	−0.02 [−0.03, −0.01]	.003
Physical activity²	−0.02 [−0.02, −0.01]	< .001	0.00 [0.00, 0.01]	.612
Physical activity × Age	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity²	0.00 [−0.01, 0.00]	.001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [−0.01, 0.31]	.062	0.15 [−0.02, 0.32]	.086
Physical activity	0.03 [0.03, 0.04]	< .001	0.02 [0.02, 0.03]	< .001
Age	−0.01 [−0.02, 0.01]	.323	−0.01 [−0.03, 0.00]	.067
Physical activity²	−0.02 [−0.02, −0.01]	< .001	−0.02 [−0.02, −0.01]	< .001
Physical activity × Age	−0.02 [−0.02, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Physical activity²	0.00 [0.00, 0.00]	.095	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.09 [−0.18, 0.37]	.503	0.12 [−0.14, 0.37]	.379
Physical activity	−0.12 [−0.12, −0.11]	< .001	−0.06 [−0.06, −0.05]	< .001
Age	−0.01 [−0.02, 0.00]	.036	0.00 [−0.02, 0.01]	.524
Physical activity²	0.02 [0.02, 0.03]	< .001	−0.01 [−0.02, −0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Age	−0.01 [−0.01, −0.01]	< .001	0.01 [0.00, 0.01]	< .001
Age × Physical activity²	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.18 [0.04, 0.33]	.014	0.06 [−0.14, 0.25]	.569
Physical activity	0.29 [0.27, 0.30]	< .001	0.14 [0.13, 0.15]	< .001
Age	0.00 [−0.02, 0.02]	.730	−0.08 [−0.10, −0.06]	< .001
Physical activity²	−0.15 [−0.16, −0.14]	< .001	−0.09 [−0.09, −0.08]	< .001
Physical activity × Age	0.00 [−0.01, 0.00]	.049	−0.05 [−0.05, −0.05]	< .001
Age × Physical activity²	−0.03 [−0.04, −0.03]	< .001	0.01 [0.01, 0.01]	< .001

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

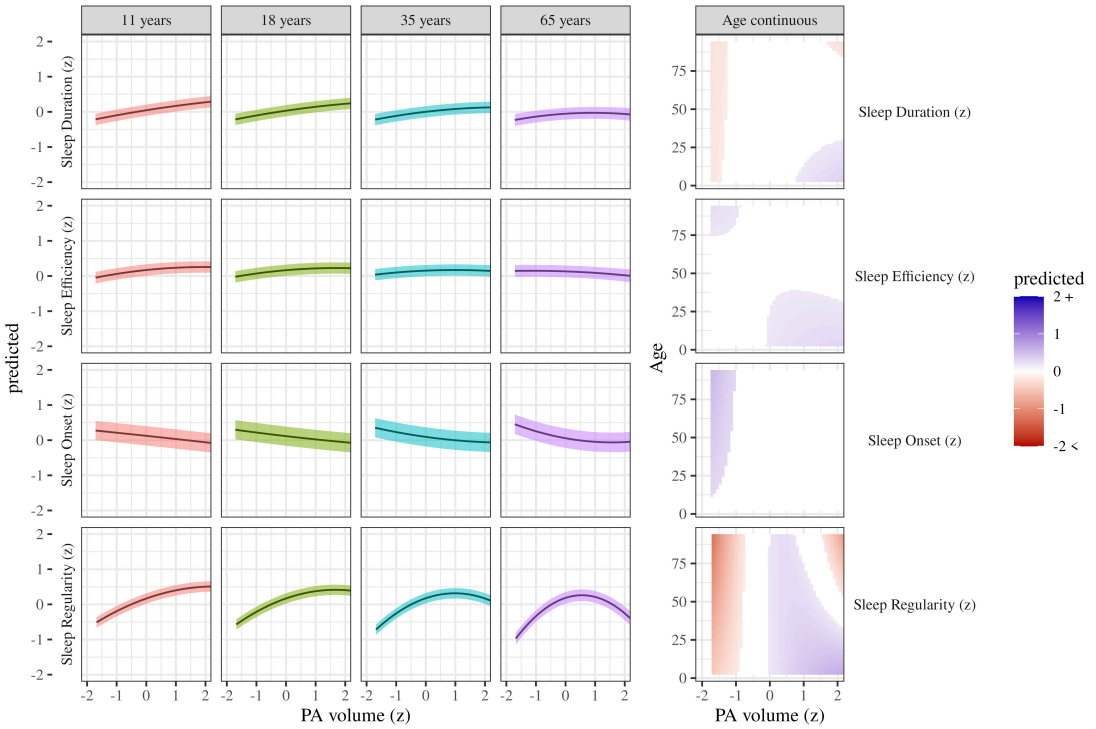


Figure 1: Standardized sleep metrics predicted by physical activity volume. The panels on the left show the curvilinear relationship between PA volume and each of the sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

Physical activity intensity and sleep

Physical activity intensity was weakly associated with sleep duration, efficiency, and onset, and more strongly with sleep regularity (Table 2; Figure 2). Curvilinear terms were statistically significant for all sleep outcomes except duration but were meaningful only for sleep regularity. Lower intensity was associated with poorer regularity, and the benefit of higher intensity diminished beyond 1 SD above the mean. This association was steepest in children and young adults, although few of them had very low intensity. The associations with sleep duration and efficiency weakened with age, and the association with sleep regularity declined markedly, from $\beta = 0.26$ [0.25, 0.28] at age 11 to 0.00 [−0.02, 0.01] at age 65.

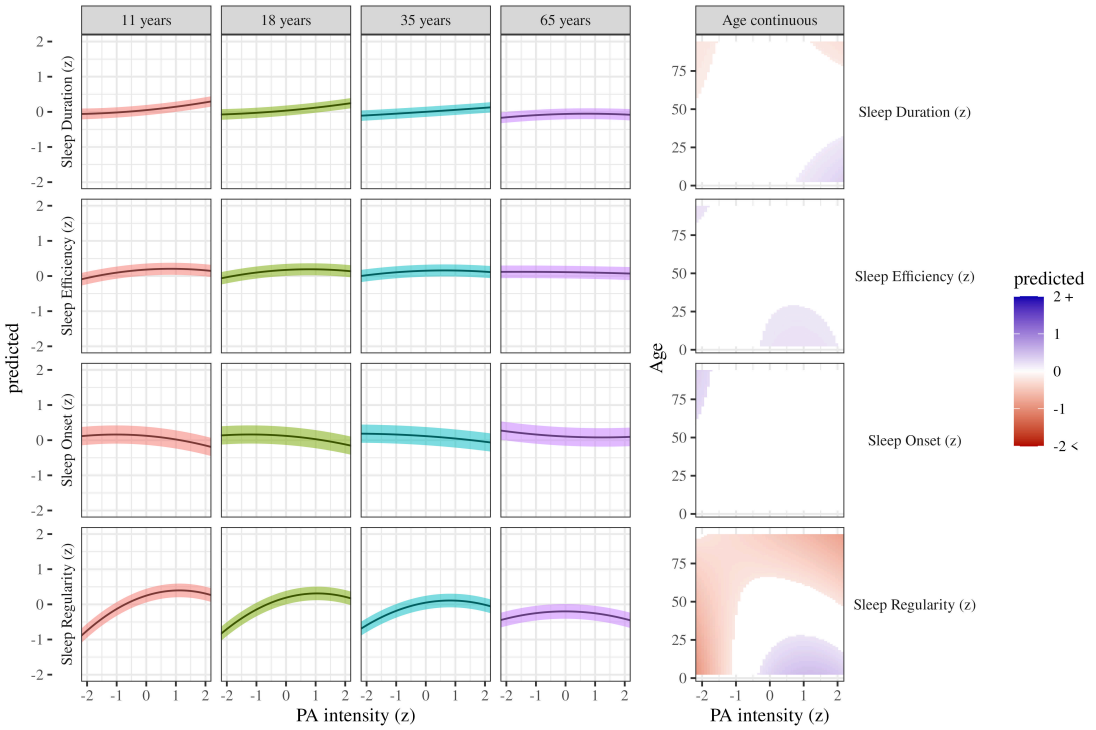


Figure 2: Sleep metrics predicted by physical activity intensity. The panels on the left show the curvilinear relationship between PA intensity and sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

Sleep and next-day physical activity

There was a weak curvilinear association between sleep duration and next-day physical activity volume and intensity, with physical activity highest after nights of slightly longer than average sleep (Table 3; Figure 3). The association with volume weakened with age and was slightly negative by age 65 (β = 0.05 [0.04, 0.05] at age 11 and −0.02 [−0.02, −0.01] at age 65), whereas the association with intensity changed little with age. Sleep efficiency had a negligible association with next-day volume at all ages (β = −0.01 [−0.01, 0.00] at age 11 and 0.01 [0.01, 0.02] at age 65) and only a weak positive association with intensity (Figure 4). Earlier sleep onset was weakly associated with higher next-day volume and intensity, and these associations were attenuated at older ages (volume: β = −0.03 [−0.04, −0.02] at age 11 and −0.01 [−0.02, −0.01] at age 65; Figure 5). By age 65, earlier onset was associated with marginally lower next-day intensity (β = 0.01 [0.00, 0.02]). More regular sleep was linearly associated with higher next-day volume and intensity (Figure 6). These associations weakened with age (volume: β = 0.14 [0.13, 0.14] at age 11 and 0.09 [0.08, 0.10] at age 65), and the association with intensity was negligible in older adults (β = 0.02 [0.02, 0.03] at age 65).

Table 3: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [−0.17, 0.44]	.378	−0.07 [−0.19, 0.05]	.254
Sleep duration	0.02 [0.01, 0.02]	< .001	0.03 [0.02, 0.03]	< .001
Age	−0.23 [−0.24, −0.22]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration²	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep duration × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.051
Age × Sleep duration²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.326
Sleep efficiency (z)				
(Intercept)	0.13 [−0.17, 0.42]	.412	−0.07 [−0.20, 0.05]	.242
Sleep efficiency	0.00 [0.00, 0.01]	.279	0.01 [0.00, 0.01]	.009
Age	−0.23 [−0.24, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep efficiency²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.200
Age × Sleep efficiency²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.227
Sleep onset (z)				
(Intercept)	0.13 [−0.18, 0.43]	.410	−0.08 [−0.21, 0.04]	.202
Sleep onset	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Age	−0.22 [−0.23, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset²	0.00 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.811

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset × Age	0.00 [0.00, 0.00]	.012	0.01 [0.01, 0.01]	< .001
Age × Sleep onset²	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.020
Sleep regularity (z)				
(Intercept)	0.04 [−0.27, 0.34]	.817	−0.18 [−0.30, −0.05]	.006
Sleep regularity	0.12 [0.11, 0.12]	< .001	0.06 [0.06, 0.07]	< .001
Age	−0.24 [−0.26, −0.22]	< .001	−0.32 [−0.34, −0.31]	< .001
Sleep regularity²	0.00 [0.00, 0.00]	.448	0.00 [0.00, 0.00]	.414
Sleep regularity × Age	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Sleep regularity²	0.00 [0.00, 0.00]	.145	0.00 [0.00, 0.00]	.023

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

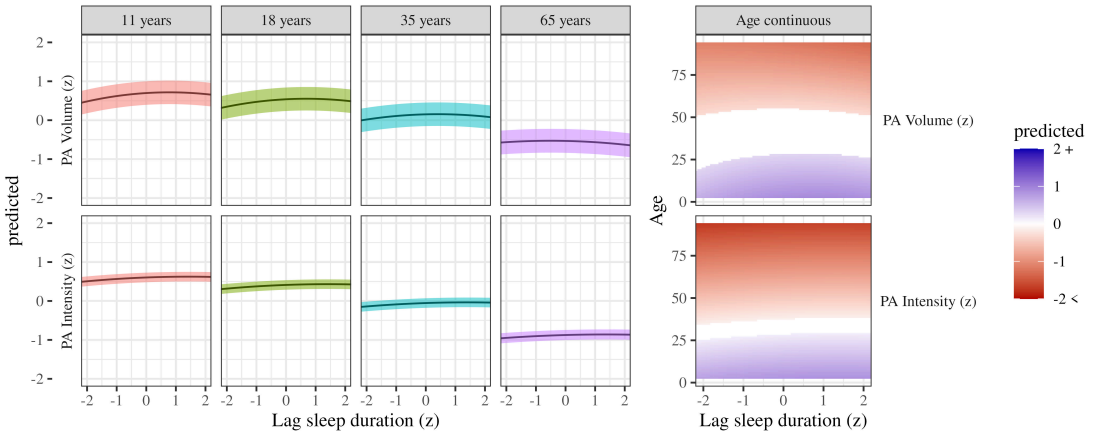


Figure 3: Physical activity is predicted by the duration of sleep the previous night. The panels on the left show the curvilinear relationship between sleep duration and the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

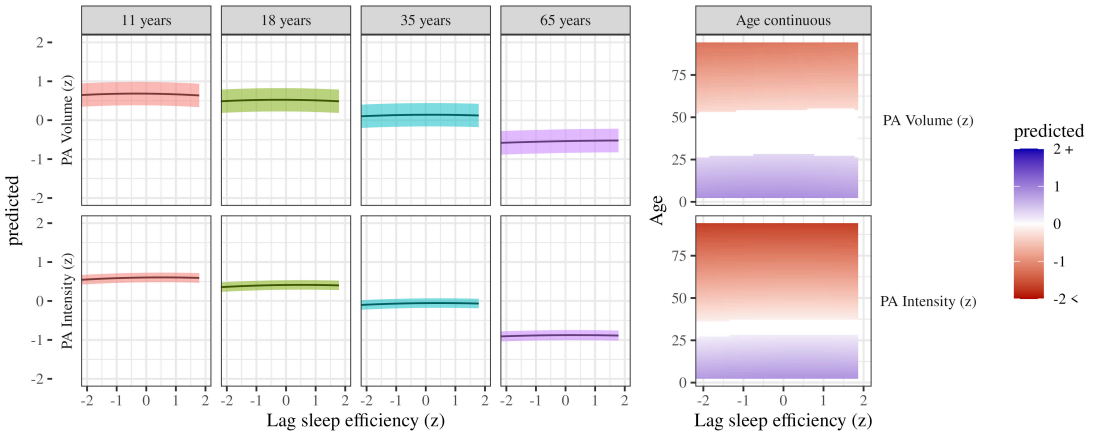


Figure 4: Physical activity predicted by previous night sleep efficiency. The panels on the left show the curvilinear relationship between sleep efficiency and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

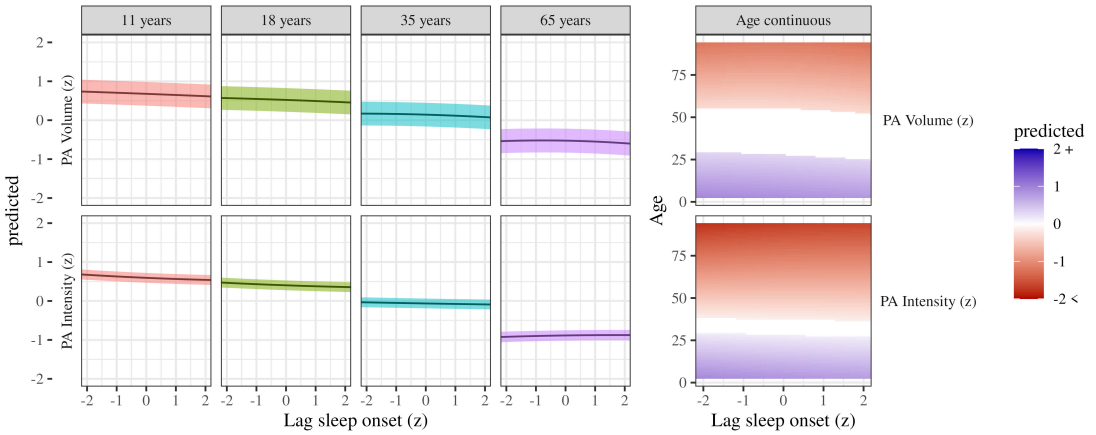


Figure 5: Physical activity predicted by previous night sleep onset. The panels on the left show the curvilinear relationship between sleep onset and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

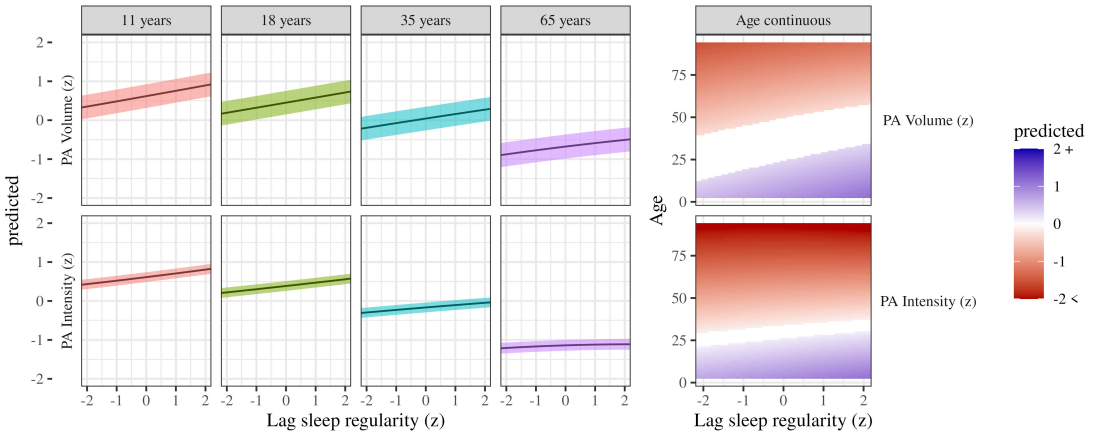


Figure 6: Physical activity predicted by previous night sleep regularity. The panels on the left show the curvilinear relationship between sleep regularity and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

Discussion

Pooling accelerometer data from 20 studies spanning the lifespan, we found that both the volume and intensity of physical activity were associated with longer sleep duration, higher sleep efficiency, earlier sleep onset, and more regular sleep. These relationships were the strongest for sleep regularity, although the relationship with physical activity intensity weakened with age. Although robust, these associations were small: a 1 SD (23.7 mg) higher physical activity volume was associated with 4.7 points higher sleep regularity, 11.8 min longer sleep, and 11 min earlier sleep onset at age 11, and with 4.4 points, 4.8 min, and 19 min, respectively, at age 65.

In the reverse direction, more regular sleep was followed by more total and more intense physical activity the next day, although this association weakened with age, whereas associations of

sleep onset, duration, and efficiency with next-day physical activity were small. Our sensitivity analyses suggest that these findings were largely robust to the decisions made during analysis.

Our findings broadly align with meta-analyses of experimental research, although the strengths of the relationships we observed were much weaker.[13,14] For example, Kredlow and colleagues[13] found small ($0.22 - 0.30$) but significant relationships between physical activity and sleep duration and efficiency in adults using experimental studies for both acute and regular physical activity. These effect sizes are not directly comparable, however, as the experiments manipulated physical activity, whereas we examined habitual physical activity. Accelerometry also distinguishes quiet wakefulness from sleep less well than polysomnography, particularly in children,[33,34] which may attenuate associations. Trials also suggest that the benefits of exercise for sleep are concentrated in people whose sleep is poor at baseline,[14,15] whereas our analysis did not target people with poor sleep. The timing of activity may also matter, as strenuous exercise close to bedtime can delay and disrupt sleep,[43,44] although in our data associations were broadly similar regardless of the time of day of the most active hour, and across seasons.

Among observational studies of day-to-day sleep and physical activity, a meta-analysis of 33 studies in adults did not support a bidirectional daily association.[21] Some sleep parameters were weakly associated with next-day physical activity, and more active individuals slept slightly less. Two recent studies of consumer wearables found that previous-day activity had little association with sleep,[23] that shorter sleep was followed by more activity the next day,[23,24] and that earlier-than-usual sleep onset was followed by more moderate-to-vigorous activity.[24] Our finding that earlier sleep onset preceded more physical activity is consistent with these studies, although the association was small in our data. However, we found clearer associations between physical activity and that night's sleep, and physical activity was highest after slightly longer, rather than shorter, than average sleep. One explanation is that our models did not separate within-person (day-to-day) from between-person (habitual) associations, which can differ,[45] so our estimates partly reflect stable differences between people.

The association between sleep regularity and physical activity appeared bidirectional. Physical activity predicted more regular sleep that night, and more regular sleep predicted more physical activity the next day. This is consistent with the idea that sleep and physical activity are mutually reinforcing behaviours,[12] or that both reflect shared underlying factors, such as daily routines, work or school schedules, and health priorities. Consistent physical activity may also help stabilise circadian rhythms.[25] Sleep regularity is a stronger predictor of mortality than sleep duration,[46] and irregular sleepers remain at higher risk of cardiovascular events and depression even when they meet sleep duration recommendations,[47,48] so the finding that physical activity is associated with sleep regularity may be particularly important.

The association between physical activity volume and sleep regularity was largely consistent across the lifespan, as a large study of adult wearable users found for other day-level associations.[23] Other associations changed more markedly with age: those of physical activity intensity with sleep regularity, and of sleep with next-day physical activity, weakened with age and were small or negligible in older adults, consistent with evidence that the interplay between sleep and activity differs in very old adults.[45] That our strongest association varied little with age suggests that the clearer evidence from meta-analyses in adults than in children and adolescents[12,13,17,18] may reflect study design or measurement rather than age.

Strengths of our study include device-measured sleep and physical activity, harmonised from raw data collected with 24-hour wear protocols across multiple studies, regions, and age groups, and the avoidance of intensity cut-points. It also has limitations. Despite our efforts to harmonise data, variability in accelerometer wear locations and slight differences in data collection protocols across studies may have introduced measurement inconsistencies. Physical activity estimates from different wear locations are strongly correlated but agree less well in absolute terms, particularly between wrist and hip.[49] To maximise the data available, we included participants with four or more days of valid wear time, although some recommend at least seven nights to estimate sleep regularity[50] and reliable estimates of night-to-night variability in sleep may require more than a month of data;[51] restricting the sample to participants with at least seven valid days gave a similar association between physical activity volume and sleep regularity (β = 0.31 [0.30, 0.33] at age 11 and 0.24 [0.22, 0.27] at age 65). As in any observational study, the temporal ordering of our lagged models does not establish causality, and although we controlled for several potential confounders, unmeasured factors may explain the observed associations. Finally, adolescents and young adults were under-represented, which limits generalisability to these groups and matters because young adults are the most likely to report difficulty falling asleep.[8]

These findings have implications for public health recommendations. Our results support promoting physical activity as a strategy for maintaining regular sleep patterns across the lifespan. This message fits with Australia's recent 24-hour movement guidelines for adults, which pair physical activity recommendations with consistent bed and wake times.[52] Given the well-established links between sleep and overall health,[3,4] incorporating physical activity into public health messaging may have downstream benefits for population health. However, physical activity was more weakly and less consistently associated with sleep duration, efficiency, and onset, so public health messages should emphasise its role in promoting consistent sleep schedules rather than framing it as a universal solution for improving sleep. We did not examine these relationships specifically in individuals with sleep disorders or other health conditions, where the effects of physical activity may differ. Future research should test whether physical activity interventions improve sleep regularity, whether more regular sleep in turn supports physical activity, and whether either brings health benefits beyond sleep itself.

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